

RFF-SP emissions

*10,000-member probabilistic
socioeconomic + emissions ensemble*



FaIR v2.2.4 climate model

*841 calibrated configs × stochastic seeds
→ probabilistic GMST(t) + OHC(t)*



MimiBRICK SLR + Wong importance weighting

*10,000-member BRICK posterior;
Wong (2025) AR(1) likelihood vs. observed GMSL*



Probabilistic GMSL(t) — cm rel. year 2000

*13,500-tuple 4-way H-S decomp; 2020–2300 horizon
(emissions / climate cfg / internal var / BRICK)*



FrEDI damage-function translation

*11-yr rolling avg × interpolation between 6 Sweet et al. (2022)
NCA5 nodes: 30, 50, 100, 150, 200, 300 cm GMSL*



State-level economic + physical damages

Coastal Protection — HFE Transportation — HFE Mortality (Shanghai)